

ENGR 213 · Gradesaver Master Solving System & Exam Blueprint

Complete Analytical Framework, Step-by-Step Solution Protocols & Exam Problem Bank

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Executive Overview: The Gradesaver Methodology

The **Gradesaver System** is a methodical, step-by-step problem-solving architecture developed to eliminate ambiguity and mechanical errors on ENGR 213 quizzes, midterms, and final examinations. Differential equations are frequently perceived as a collection of disjointed tricks; in reality, every first-order differential equation can be mapped into a deterministic decision matrix.

The 5-Phase Universal Solution Protocol

Every single problem on the examination must be processed through these five distinct operational phases:



- Phase 1: Standard Normalization:** Isolate $\frac{dy}{dx}$ with unit coefficient ($1 \cdot y'$), or express in symmetric differential form $M(x, y), dx + N(x, y), dy = 0$.
- Phase 2: Diagnostic Form Test:** Run the 4-Gate Classifier:
 - Gate A:* Can variables be separated as $h(y), dy = g(x), dx$?
 - Gate B:* Is it linear in $y, y' + P(x)y = f(x)$? (Or linear in $x, \frac{dx}{dy} + P(y)x = f(y)$?)
 - Gate C:* Is it exact, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$? If not, does a 1-variable integrating factor $\mu(x)$ or $\mu(y)$ exist?
 - Gate D:* Is it reducible by substitution? ($u = Ax + By + C$, Homogeneous $y = ux$, or Bernoulli $u = y^{1-n}$).
- Phase 3: Transformation & Analytical Integration:** Apply the designated transformation, compute antiderivatives with exact constants $+C$.
- Phase 4: Parameter Determination (C):** If an Initial Value Problem (IVP) is given ($y(x_0) = y_0$), substitute immediately to find the unique constant C .
- Phase 5: Interval of Validity (I) & Singularities:** State the maximal open interval containing x_0 where the solution and its derivative are real, finite, and continuous.

Section 1: Separable Differential Equations ($y' = g(x)h(y)$)

1.1 The Tutor Heuristic ("The Move Here")

A first-order equation is separable if all y -dependencies can be factored onto the differential dy and all x -dependencies onto dx .

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)}, dy = g(x), dx$$

Critical Watch-Out (Equilibrium Singular Solutions):

Before dividing by $h(y)$, determine all roots y^* such that $h(y^*) = 0$. These constant functions $y(x) = y^*$ are **equilibrium solutions**. If an equilibrium solution cannot be obtained by choosing any value of the integration constant C , it is a **singular solution**.

1.2 Step-by-Step Exam Protocol

- Step 1 (Factorization):** Rewrite into $M(x), dx + N(y), dy = 0$.
- Step 2 (Division & Separation):** Divide by $N(y)M(x)$ and set up integrals:

$$\int \frac{1}{h(y)}, dy = \int g(x), dx + C$$

- Step 3 (Integration & Logarithmic Exponentiation):** When integrals produce $\ln |y|$, exponentiate:

$$|y| = e^{\int g(x)dx + C} = e^C e^{\int g(x)dx} = K e^{\int g(x)dx}, \quad K = \pm e^C \neq 0$$

Allowing $K = 0$ frequently absorbs the equilibrium solution $y = 0$.

- Step 4 (IVP Substitution):** Plug in x_0, y_0 before complicating algebraic inversion.
- Step 5 (Explicit Branch Selection):** If $y^2 = f(x)$, select the sign $\pm \sqrt{f(x)}$ matching $y(x_0) = y_0$.

1.3 Exemplar Problems from Tutor Blueprint

Problem 1.1: Separable Trigonometric IVP

$$\frac{dy}{dx} = 3x^2(y^2 + 1), \quad y(0) = 1$$

Solution Walkthrough:

1. **Identify and Separate:**

$$\frac{dy}{y^2 + 1} = 3x^2, dx$$

2. **Integrate both sides:**

$$\int \frac{dy}{y^2 + 1} = \int 3x^2, dx \implies \arctan(y) = x^3 + C$$

3. **Apply Initial Condition** $y(0) = 1$:

$$\arctan(1) = 0^3 + C \implies C = \frac{\pi}{4}$$

4. **Isolate y explicitly:**

$$y(x) = \tan\left(x^3 + \frac{\pi}{4}\right)$$

5. **Interval of Validity:** The tangent function has vertical asymptotes at $\pm \frac{\pi}{2}$. Therefore:

$$-\frac{\pi}{2} < x^3 + \frac{\pi}{4} < \frac{\pi}{2} \implies -\frac{3\pi}{4} < x^3 < \frac{\pi}{4} \implies \left(-\frac{3\pi}{4}\right)^{1/3} < x < \left(\frac{\pi}{4}\right)^{1/3}$$

$$\text{Interval of definition: } I = \left(-\left(\frac{3\pi}{4}\right)^{1/3}, \left(\frac{\pi}{4}\right)^{1/3}\right).$$

Section 2: First-Order Linear Equations ($y' + P(x)y = f(x)$)

2.1 The Integrating Factor Framework

For any linear first-order ODE:

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

2.2 Step-by-Step Exam Protocol

- **Step 1 (Normalization):** Divide by $a_1(x)$ to establish the standard canonical form:

$$\frac{dy}{dx} + P(x)y = f(x), \quad P(x) = \frac{a_0(x)}{a_1(x)}, \quad f(x) = \frac{g(x)}{a_1(x)}$$

Singularity Audit: Singularities occur wherever $a_1(x) = 0$.

- **Step 2 (Integrating Factor Computation):**

$$I(x) = e^{\int P(x), dx}$$

Rule: Do not add a constant $+C$ to the exponent of $I(x)$. Simplify $e^{n \ln |x|} = |x|^n \implies x^n$.

- **Step 3 (Product Rule Collapse):**

$$\frac{d}{dx} [I(x)y] = I(x)f(x)$$

- **Step 4 (Integration):**

$$I(x)y = \int I(x)f(x), dx + C \implies y(x) = \frac{1}{I(x)} \left(\int I(x)f(x), dx + C \right)$$

- **Step 5 (Inversion & Interval of Definition):** Interval I is the continuous segment around x_0 where both $P(x)$ and $f(x)$ are continuous.

2.3 Exemplar Problems from Tutor Blueprint

Problem 2.1: Singular Linear ODE

$$x \frac{dy}{dx} - 3y = x^3, \quad x > 0$$

Solution Walkthrough:

1. **Normalize:**

$$y' - \frac{3}{x}y = x^2$$

2. **Integrating Factor:**

$$P(x) = -\frac{3}{x} \implies \int P(x), dx = -3 \ln(x) = \ln(x^{-3})$$

$$I(x) = e^{\ln(x^{-3})} = x^{-3} = \frac{1}{x^3}$$

3. **Multiply and Collapse:**

$$\frac{d}{dx} [x^{-3}y] = x^{-3} \cdot x^2 = \frac{1}{x}$$

4. **Integrate:**

$$x^{-3}y = \int \frac{1}{x}, dx + C = \ln|x| + C$$

5. **General Solution** ($x > 0$):

$$y(x) = x^3(\ln(x) + C)$$

Problem 2.2: Trigonometric Linear IVP

$$y' + y \tan(x) = \cos^2(x), \quad y(0) = -1$$

Solution Walkthrough:

1. **Identify** $P(x) = \tan(x)$ and $f(x) = \cos^2(x)$:

2. **Integrating Factor:**

$$I(x) = e^{\int \tan(x) dx} = e^{\ln|\sec(x)|} = \sec(x)$$

3. **Collapse and Integrate:**

$$\begin{aligned} \frac{d}{dx} [\sec(x)y] &= \sec(x) \cos^2(x) = \cos(x) \\ \sec(x)y &= \int \cos(x) dx + C = \sin(x) + C \end{aligned}$$

4. **General Solution:**

$$y(x) = \frac{\sin(x) + C}{\sec(x)} = (\sin(x) + C) \cos(x)$$

5. **Apply Initial Condition** $y(0) = -1$:

$$-1 = (0 + C) \cos(0) = C \implies C = -1$$

$$y(x) = (\sin(x) - 1) \cos(x) = \sin(x) \cos(x) - \cos(x) = \frac{1}{2} \sin(2x) - \cos(x)$$

6. **Interval of Validity:** $I = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Section 3: Exact Differential Equations & Integrating Factors

3.1 Test for Exactness

A differential equation written as:

$$M(x, y), dx + N(x, y), dy = 0$$

is **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

which guarantees the existence of a potential function $\Psi(x, y)$ such that $d\Psi = M, dx + N, dy = 0$, giving the implicit general solution:

$$\Psi(x, y) = C$$

3.2 Constructing $\Psi(x, y)$ (The Tutor Two-Path Method)

- **Path A (Integrating with respect to x):**

$$\Psi(x, y) = \int M(x, y), dx + g(y)$$

Differentiating with respect to y :

$$\frac{\partial \Psi}{\partial y} = \frac{\partial}{\partial y} \left(\int M(x, y), dx \right) + g'(y) = N(x, y)$$

Isolate $g'(y)$, verify that all x -terms cancel, integrate to find $g(y)$, and assemble $\Psi(x, y) = C$.

3.3 Non-Exact Equations & Special Integrating Factors

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$:

- **Case 1 (Function of x alone):**

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x) \implies \mu(x) = e^{\int f(x) dx}$$

- **Case 2 (Function of y alone):**

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = g(y) \implies \mu(y) = e^{\int g(y) dy}$$

3.4 Exemplar Problem from Tutor Blueprint

Problem 3.1: Non-Exact Equation with Integrating Factor

$$(x + y), dx + x \ln(x), dy = 0, \quad x > 1$$

Solution Walkthrough:

1. $M = x + y \implies \frac{\partial M}{\partial y} = 1$
2. $N = x \ln(x) \implies \frac{\partial N}{\partial x} = \ln(x) + 1$
3. Difference: $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 1 - (\ln(x) + 1) = -\ln(x)$
4. Divide by N :

$$\frac{-\ln(x)}{x \ln(x)} = -\frac{1}{x} = f(x)$$

5. Integrating factor:

$$\mu(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln(x)} = \frac{1}{x}$$

6. Multiply the entire ODE by $\frac{1}{x}$:

$$\left(1 + \frac{y}{x}\right) dx + \ln(x), dy = 0$$

$$\text{Now } \tilde{M} = 1 + \frac{y}{x} \implies \frac{\partial \tilde{M}}{\partial y} = \frac{1}{x}; \tilde{N} = \ln(x) \implies \frac{\partial \tilde{N}}{\partial x} = \frac{1}{x}. \text{ Exact!}$$

7. Integrate $\tilde{N}$ with respect to y :

$$\Psi(x, y) = \int \ln(x), dy = y \ln(x) + h(x)$$

$$\frac{\partial \Psi}{\partial x} = \frac{y}{x} + h'(x) = 1 + \frac{y}{x} \implies h'(x) = 1 \implies h(x) = x$$

8. General Solution:

$$\Psi(x, y) = y \ln(x) + x = C \implies y(x) = \frac{C - x}{\ln(x)}$$

Section 4: Solutions by Substitution

4.1 Type 1: Linear Combination $y' = f(Ax + By + C)$

When x and y only appear inside an expression of the form $Ax + By + C$:

- **The Move Here:** Let $u = Ax + By + C$.
- Then $\frac{du}{dx} = A + B \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{B} \left(\frac{du}{dx} - A \right)$.
- Substitution always converts the ODE into a **separable** differential equation in u and x !

Problem 4.1: Linear Combination IVP

$$y' - (4x - y + 1)^2 = 0, \quad y(0) = 1$$

(From Tutor Notes Page 32)

Solution Walkthrough:

1. **Identify Form:** $y' = (4x - y + 1)^2$. Here $A = 4, B = -1, C = 1$.

2. **Substitute:**

$$u = 4x - y + 1 \implies \frac{du}{dx} = 4 - \frac{dy}{dx} \implies \frac{dy}{dx} = 4 - \frac{du}{dx}$$

3. **Rewrite Differential Equation:**

$$4 - \frac{du}{dx} = u^2 \implies \frac{du}{dx} = 4 - u^2 = -(u^2 - 4)$$

4. **Separate Variables:**

$$\frac{du}{u^2 - 4} = -dx$$

5. **Partial Fraction Decomposition:**

$$\begin{aligned} \frac{1}{(u-2)(u+2)} &= \frac{1}{4} \left(\frac{1}{u-2} - \frac{1}{u+2} \right) \\ \frac{1}{4} \ln \left| \frac{u-2}{u+2} \right| &= -x + C_1 \implies \ln \left| \frac{u-2}{u+2} \right| = -4x + C_2 \\ \frac{u-2}{u+2} &= K e^{-4x}, \quad K = \pm e^{C_2} \end{aligned}$$

6. **Apply Initial Condition to Find K :** At $x = 0$: $y(0) = 1 \implies u(0) = 4(0) - 1 + 1 = 0$.

$$\frac{0-2}{0+2} = K e^0 \implies -1 = K \implies K = -1$$

7. **Solve for u :**

$$\begin{aligned} \frac{u-2}{u+2} &= -e^{-4x} \implies u-2 = -e^{-4x}(u+2) = -u e^{-4x} - 2e^{-4x} \\ u(1 + e^{-4x}) &= 2 - 2e^{-4x} \implies u(x) = \frac{2(1 - e^{-4x})}{1 + e^{-4x}} = 2 \tanh(2x) \end{aligned}$$

8. **Back-Substitute $u = 4x - y + 1$ to Find $y(x)$:**

$$4x - y + 1 = 2 \tanh(2x) \implies y(x) = 4x + 1 - 2 \tanh(2x)$$

(Note: $\tanh(0) = 0 \implies y(0) = 1$, verified perfectly!)

4.2 Type 2: Homogeneous Differential Equations (y/x)

If $M(tx, ty) = t^k M(x, y)$ and $N(tx, ty) = t^k N(x, y)$:

- **The Move Here:** Let $y = ux \implies \frac{dy}{dx} = u + x \frac{du}{dx}$.
- The equation transforms into a separable ODE in u and x .

Problem 4.2: Homogeneous Algebraic Equation

$$(x - y), dx + x, dy = 0$$

(From Tutor Notes Page 23)

Solution Walkthrough:

1. **Check Homogeneity:** $M(tx, ty) = t(x - y)$ and $N(tx, ty) = tx$. Degree is 1.

2. **Substitute $y = ux \implies dy = u, dx + x, du$:**

$$\begin{aligned} (x - ux), dx + x(u, dx + x, du) &= 0 \\ x(1 - u), dx + xu, dx + x^2, du &= 0 \\ x, dx - xu, dx + xu, dx + x^2, du &= 0 \implies x, dx + x^2, du = 0 \end{aligned}$$

3. **Separate Variables ($x \neq 0$):**

$$\frac{1}{x}, dx + du = 0 \implies \int du = - \int \frac{1}{x}, dx \implies u = -\ln|x| + C$$

4. **Back-Substitute $u = \frac{y}{x}$:**

$$\frac{y}{x} = C - \ln|x| \implies y(x) = x(C - \ln|x|)$$

4.3 Type 3: Bernoulli Equations ($y' + P(x)y = f(x)y^n$)

- **The Move Here:** Divide by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = f(x)$$

- Set $u = y^{1-n} \implies \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx} \implies y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{du}{dx}$.
- Substitute to obtain a linear 1st-order equation:

$$\frac{du}{dx} + (1-n)P(x)u = (1-n)f(x)$$

Problem 4.3: Bernoulli Equation from Tutor Notes

$$x \frac{dy}{dx} - (1+x)y = xy^2, \quad x > 0$$

(From Tutor Notes Page 28)

Solution Walkthrough:

1. **Normalize:**

$$y' - \frac{1+x}{x}y = y^2 \implies n = 2$$

2. **Divide by y^2 :**

$$y^{-2}y' - \left(\frac{1}{x} + 1\right)y^{-1} = 1$$

3. **Substitute $u = y^{1-2} = y^{-1} \implies u' = -y^{-2}y'$:**

$$-u' - \left(\frac{1}{x} + 1\right)u = 1 \implies u' + \left(\frac{1}{x} + 1\right)u = -1$$

4. **Integrating Factor:**

$$I(x) = e^{\int (1/x+1)dx} = e^{\ln(x)+x} = xe^x$$

5. **Collapse and Integrate:**

$$\begin{aligned} \frac{d}{dx}[xe^xu] &= -xe^x \\ xe^xu &= -\int xe^x dx = -(xe^x - e^x) + C = -xe^x + e^x + C \end{aligned}$$

6. **Solve for u :**

$$u(x) = -1 + \frac{1}{x} + \frac{C}{xe^x} = \frac{1-x+Ce^{-x}}{x}$$

7. **Back-Substitute $y = \frac{1}{u}$:**

$$y(x) = \frac{x}{1-x+Ce^{-x}}$$

Section 5: Engineering Applications & Forensic Modeling

5.1 Newton's Law of Cooling / Heating ($dT/dt = k(T - T_m)$)

$$\frac{dT}{dt} = k(T - T_m) \implies T(t) = T_m + (T_0 - T_m)e^{kt}$$

(Note: $k < 0$ for natural heat transfer toward medium temperature T_m).

Problem 5.1: The 3-Point Forensic Oven Problem

Problem Statement (Tutor Notes Page 35):

A thermometer reading 70°F is placed in an oven preheated to an unknown constant temperature T_m . An observer through the oven window records that the thermometer reads 110°F after 0.5 min, and 145°F after 1 min. Determine the temperature of the oven T_m .

Solution Walkthrough:

1. **Model:**

$$T(t) = T_m + Ce^{kt}$$

2. **Point 1** ($t = 0, T = 70$):

$$70 = T_m + C \implies C = 70 - T_m$$

$$T(t) = T_m + (70 - T_m)e^{kt}$$

3. **Point 2** ($t = 0.5, T = 110$):

$$110 - T_m = (70 - T_m)e^{0.5k} \implies e^{0.5k} = \frac{110 - T_m}{70 - T_m}$$

4. **Point 3** ($t = 1.0, T = 145$):

$$145 - T_m = (70 - T_m)e^k = (70 - T_m)(e^{0.5k})^2$$

5. **Substitute** $(e^{0.5k})^2$:

$$145 - T_m = (70 - T_m) \left(\frac{110 - T_m}{70 - T_m} \right)^2 = \frac{(110 - T_m)^2}{70 - T_m}$$

6. **Cross-Multiply and Expand:**

$$(145 - T_m)(70 - T_m) = (110 - T_m)^2$$

$$10150 - 215T_m + T_m^2 = 12100 - 220T_m + T_m^2$$

7. **Cancel T_m^2 and Solve for T_m :**

$$-215T_m + 220T_m = 12100 - 10150$$

$$5T_m = 1950 \implies T_m = 390^\circ\text{F}$$

The oven temperature is exactly 390°F .

5.2 Mixture Tanks

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}} = (C_{\text{in}} \cdot r_{\text{in}}) - \left(\frac{A(t)}{V(t)} \cdot r_{\text{out}} \right)$$

Where $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$.

Problem 5.2: Evacuating Tank with Decreasing Volume

Problem Statement (Tutor Notes Page 39):

A tank initially holds 100 gal of pure water. Brine containing 3 lb/gal of salt enters at 4 gal/min, and the well-stirred mixture leaves at 5 gal/min. Find the amount of salt in the tank after 30 min.

Solution Walkthrough:

1. **Flow Rates & Volume:**

$$r_{\text{in}} = 4, \quad r_{\text{out}} = 5 \implies \Delta r = -1 \implies V(t) = 100 - t$$

Tank empties at $t = 100$ min.

2. **Rates:**

$$R_{\text{in}} = 3 \times 4 = 12 \text{ lb/min}$$

$$R_{\text{out}} = \frac{A(t)}{100 - t} \times 5 = \frac{5}{100 - t} A(t)$$

3. **Differential Equation:**

$$\frac{dA}{dt} + \frac{5}{100 - t} A = 12, \quad A(0) = 0$$

4. **Integrating Factor:**

$$P(t) = \frac{5}{100 - t} \implies \int P(t) dt = -5 \ln(100 - t) = \ln((100 - t)^{-5})$$

$$I(t) = (100 - t)^{-5}$$

5. **Integrate:**

$$\frac{d}{dt} [(100 - t)^{-5} A] = 12(100 - t)^{-5}$$

$$(100 - t)^{-5} A = 12 \int (100 - t)^{-5} dt = 12 \cdot \frac{(100 - t)^{-4}}{(-4)(-1)} + C = 3(100 - t)^{-4} + C$$

$$A(t) = 3(100 - t) + C(100 - t)^5$$

6. **Initial Condition** $A(0) = 0$:

$$0 = 3(100) + C(100)^5 \implies C = -\frac{300}{100^5} = -\frac{3}{100^4} = -3 \times 10^{-8}$$

7. **Evaluate at** $t = 30$ min:

$$100 - 30 = 70$$

$$A(30) = 3(70) - 3 \times 10^{-8}(70)^5 = 210 - 3 \times 10^{-8}(1680700000) = 210 - 50.42 = 159.58 \text{ lbs}$$

Amount of salt in tank at $t = 30$ min is approximately 159.6 lbs.

5.3 First-Order LR Electric Circuits

$$L \frac{di}{dt} + Ri = E(t)$$

Problem 5.3: Constant Voltage LR Circuit

Problem Statement (Tutor Notes Page 40):

An inductance $L = 20 \text{ H}$ and resistance $R = 2, \Omega$ are connected in series with an electromotive force $E(t) = 120 \text{ V}$. Find $i(t)$ given initial current $i(0) = 0 \text{ A}$, and find $i(2)$.

Solution Walkthrough:

1. **Model:**

$$20 \frac{di}{dt} + 2i = 120 \implies \frac{di}{dt} + 0.1i = 6$$

2. **Integrating Factor:** $I(t) = e^{0.1t}$.

$$\frac{d}{dt} [e^{0.1t} i] = 6e^{0.1t} \implies e^{0.1t} i = \frac{6}{0.1} e^{0.1t} + C = 60e^{0.1t} + C$$

$$i(t) = 60 + Ce^{-0.1t}$$

3. **Apply** $i(0) = 0$:

$$0 = 60 + C \implies C = -60$$

$$i(t) = 60(1 - e^{-0.1t}) \text{ A}$$

4. **Current at** $t = 2$ s:

$$i(2) = 60(1 - e^{-0.2}) = 60(1 - 0.81873) = 10.88 \text{ A}$$

Steady-state current: $\lim_{t \rightarrow \infty} i(t) = 60 \text{ A}$.

5.4 Radioactive Decay Kinetics

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt}$$

Problem 5.4: Half-Life & Decay

Problem Statement (Tutor Notes Page 42):

Initially 100 mg of a radioactive substance is present. After 6 hours, the mass has decreased by 3. Find the mass remaining after 24 hours.

Solution Walkthrough:

1. **Model:** $P(t) = 100e^{kt}$.

2. **At** $t = 6$, **mass decreased by** 3:

$$97 = 100e^{6k} \implies e^{6k} = 0.97 \implies k = \frac{\ln(0.97)}{6} \approx -0.005076 \text{ h}^{-1}$$

3. **Amount after** $t = 24$ hours: Notice $24 = 4 \times 6$. Therefore:

$$P(24) = 100e^{24k} = 100(e^{6k})^4 = 100(0.97)^4$$

$$0.97^4 = 0.88529$$

$$P(24) = 88.53 \text{ mg}$$

Section 6: Master Diagnostic Decision Matrix

Differential Equation Form	Diagnostic Fingerprint	The Move Here	Core Formula / Transform
Separable	Products $f(x)g(y)$	Separate variables	$\int \frac{dy}{g(y)} = \int f(x)dx + C$
Linear 1st Order	Single y with no powers	Integrating Factor	$I(x) = e^{\int P(x)dx}$, $y = \frac{1}{I} \int I f dx$
Exact	$Mdx + Ndy = 0$	Cross-partial test $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$	$\Psi = \int Mdx + g(y) = C$
Non-Exact	Cross-partials unequal	Compute $\frac{M_y - N_x}{N}$ or $\frac{N_x - M_y}{M}$	$\mu(x) = e^{\int f(x)dx}$ or $\mu(y) = e^{\int g(y)dy}$
Type 1 Linear	$y' = f(Ax + By + C)$	Substitute argument	$u = Ax + By + C \implies \frac{du}{dx} = A + By'$
Homogeneous	M, N same polynomial degree	Substitute ratio	$y = ux \implies dy = udx + xdu$
Bernoulli	$y' + P(x)y = f(x)y^n$	Divide by y^n , substitute	$u = y^{1-n} \implies u' + (1-n)Pu = (1-n)f$
Newton Cooling	Heat exchange with medium	Separation of variables	$T(t) = T_m + (T_0 - T_m)e^{kt}$
Mixture Tank	Salt in liquid volume	Net rate equation	$\frac{dA}{dt} + \frac{r_{out}}{V(t)}A = C_{in}r_{in}$
LR Circuit	Inductance & resistance	Linear 1st order	$i(t) = \frac{E}{R} + \left(i_0 - \frac{E}{R}\right)e^{-Rt/L}$